Zirui Peng

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Education

B.S. Sept 2024 – present
Peking University, Physics; Double Major in Artificial Intelligence Beijing, China

- GPA 3.86/4.00 (Top 10% of the cohort); Credit-weighted score 92.3/100
- 2024 School-level Excellent Student; Recipient of the 2024 Qin Wanshun Jin Yunhui scholarship
- Selected coursework (Math and Physics): Theoretical Mechanics (98), Linear Algebra (98), Advanced Mathematics I (96), Electromagnetism (96), Methods of Mathematical Physics II (95), Electrodynamics (94), Thermal Physics (91), Optics (90), General Physics Lab I (90), Fundamentals of Theoretical Physics I (90), General Physics Lab II (89), Modern Physics (89), Mechanics (89)
- Selected coursework (CS/AI): Data Structure and Algorithm (95), Introduction to Computation (93), Machine Learning (91), Information Theory (88)

Research Internship July 2026 – Aug 2026
Massachusetts Institute of Technology, Department of Physics Cambridge, MA
Conducted research under the supervision of Prof. Jeff Gore

Main Projects

Emergent Inverse-Depth Scaling From Nonlinearity In Attention July 2026 – Sept 2026
Paper under review at ICLR | **arXiv:2610.11063**
Work done during an internship at MIT

- Found that nonlinear attention produces inverse-depth loss scaling across different data spectra by focusing directly on relevant tokens.
- Showed that nonlinear attention allows strong and weak spectral directions to be learned in parallel, whereas linear-attention models learn them sequentially.
- Linked shared error across layers to irreducible loss and layer-specific differences to continued gains with depth, drawing an analogy with the central limit theorem.

Rethinking Activation Functions: Reducing Bias Can Improve Learning Jan 2026 – Sept 2026
Paper under review at ICLR | **arXiv:2610.14500**

- Found that reducing activation bias slows initial learning but can reliably improve final performance across different activation functions.
- Linked this reversal to bias-induced overshoot, non-smooth training trajectories, transient use of bias-encoded information, and sharper local loss landscapes.
- Demonstrated consistent performance gains from bias reduction in larger vision and language models across diverse tasks.

Understanding Ensemble Averaging and Depth Utilization in Neural Networks Sept 2026 – present

- Examining how dataset size, model sparsity, and network nonlinearity affect ensemble averaging, depth utilization, and the relationship between them.
- Investigating algorithmic approaches to use model depth more efficiently, including depth mixture-of-experts architectures.

Other Contributions

Learning to Decompose Questions for Fine-Grained Evidence Selection in RAG June 2025 – Feb 2026
Paper under review at AAAI

- Identified a representative failure mode in multimodal question answering, where whole-query matching misses localized evidence.

- Developed a query decomposition and automated evidence selection pipeline to guide reasoning and answer generation.
- Optimized decomposition prompts during end-to-end training, improving performance on multimodal question-answering benchmarks.

NanoGPT Observable Explorer **observable-explorer.com** July 2026 – Aug 2026

- Built a web platform for investigating the relationship between model observables and training dynamics. Trained nanoGPT under varied experimental setups, measured a broad range of observables, and compared them across runs.

Research Interests

My research focuses on understanding neural network mechanisms and developing algorithms guided by first principles to improve model performance.

For each topic, I begin by developing a deep, well-rounded understanding of the problem, which clarifies its underlying principles and guides the design of new algorithms.

I usually develop controllable and insightful toy models that isolate individual components of larger models, allowing me to understand the key factors that shape model behavior.

My current interests are listed below, and I am open to many other research directions:

- **Neural Scaling Law**
  - Understanding how factors such as training time, model width and depth, and dataset size shape neural scaling laws.
  - Exploring algorithms that improve neural scaling efficiency and parameter utilization.
- **Optimization Dynamics**
  - Analyzing phenomena such as the edge of stability to better understand complex, high-dimensional loss landscapes.
  - Using central flow and related mathematical tools to build clear models of optimization trajectories during neural network training.
- **Generative Models**
  - Investigating whether diffusion and autoregressive models exhibit different implicit biases toward physical laws in image and video generation.
  - Developing better generative models that explicitly incorporate physical laws and constraints as inductive biases.
- **Other Topics**
  - Exploring a broader range of topics, including mechanistic interpretability, activation functions, meta-learning, and large language model systems.

Skills

Technical: Python, Markdown, LaTeX, AxMath, AxGlyph
ML Frameworks: PyTorch
Languages: Chinese (native); English (proficient) — IELTS 8.0, CET-6 641/710